

COSMIC QUEST First Round

Question Set

Version 1.0 – October 2025

Good luck

October 10, 2025

Introduction and Notes for Participants

General Instructions

- Write all numerical answers in the **units specified** in each question.
- If no units are specified, express your answer in **SI units**.
- Unless explicitly stated otherwise, assume standard physical conditions and values for universal constants.

Important Links:

- **Computational Data for Questions 26–30:** [Google Drive Link](#)
- **Clarifications and Updates Document:** [Clarifications Sheet](#)
- **Clarification Request Form:** [Clarification Request Form](#)

Physical and Astronomical Constants

Quantity	Symbol	Value (SI Units)
Speed of light in vacuum	c	$2.99792458 \times 10^8 \text{ m s}^{-1}$
Gravitational constant	G	$6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
Planck constant	h	$6.62607015 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34} \text{ J s}$
Boltzmann constant	k_B	$1.380649 \times 10^{-23} \text{ J K}^{-1}$
Avogadro constant	N_A	$6.02214076 \times 10^{23} \text{ mol}^{-1}$
Elementary charge	e	$1.602176634 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.10938356 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.6726219 \times 10^{-27} \text{ kg}$
Neutron mass	m_n	$1.674927498 \times 10^{-27} \text{ kg}$
Atomic mass unit	u	$1.66053906660 \times 10^{-27} \text{ kg}$
Stefan–Boltzmann constant	σ	$5.670374419 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Gas constant	R	$8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}$
Permittivity of free space	ϵ_0	$8.8541878128 \times 10^{-12} \text{ F m}^{-1}$
Permeability of free space	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Astronomical unit	AU	$1.495978707 \times 10^{11} \text{ m}$
Light year	ly	$9.460730472 \times 10^{15} \text{ m}$
Parsec	pc	$3.085677581 \times 10^{16} \text{ m}$
Solar mass	$M_\odot$	$1.98847 \times 10^{30} \text{ kg}$
Solar radius	$R_\odot$	$6.957 \times 10^8 \text{ m}$
Solar luminosity	$L_\odot$	$3.828 \times 10^{26} \text{ W}$
Earth mass	$M_\oplus$	$5.9722 \times 10^{24} \text{ kg}$
Earth radius	$R_\oplus$	$6.371 \times 10^6 \text{ m}$
Mean Earth–Sun distance	$a_\oplus$	1 AU
Sidereal day (Earth)	$T_\oplus$	86164 s
Mean Earth density	$\rho_\oplus$	5515 kg m^{-3}
Surface gravity (Earth)	g	9.80665 m s^{-2}
Jupiter mass	M_J	$1.89813 \times 10^{27} \text{ kg}$
Jupiter radius	R_J	$7.1492 \times 10^7 \text{ m}$
1 arcsecond	–	$4.848136811 \times 10^{-6} \text{ rad}$
1 degree	–	$\pi/180 \text{ rad}$
Planck length	l_P	$1.616255 \times 10^{-35} \text{ m}$
Planck time	t_P	$5.391247 \times 10^{-44} \text{ s}$
Planck temperature	T_P	$1.416784 \times 10^{32} \text{ K}$

1 The Wandering Star and the Forgotten Signal

Astronomers of the 43rd century report that a dim star, called *Straight Forward* in the archives, once sent Earth a strange pulse. At that time, long before the Collapse, measurements recorded its angular shuffle across the sky as **0.5"/year**, while its annual parallax was scribbled in old data chips as **0.08"**. Somewhere in the corrupted files, an entry for a scarlet shimmer in its spectrum survives: the $H\alpha$ line, not at the laboratory value of **6562.8 Å**, but strangely logged as **6561 Å**.

For 20 millennia, the star has been gliding unchanged through the void, unbothered by wars or planets, moving on as if its velocity vector were locked by some forgotten law of inertia. Historians of motion demand to know:

If one could step forward exactly **20 000 years** from that ancient measurement and reobserve the star, what would be its **radial velocity (in km/s)**?

Ignore all gravitational perturbations, galactic tides, or collapse-era debris. Work only with the values hidden in the ancient files.

2 Orbital Time Machine (An Account of the Astronomer Adruv, year 6720)

After the Collapse, Earth's astronomers rebuilt fragments of the James Webb Space Telescope (JWST) and mounted them on an orbital "time machine" above the ruins of Lagrange Point 2. According to the scattered manuals, the light-collecting surface of this hybrid relic is still about 25 m², but the onboard AI insists it now listens for "photon echoes" not only from our present epoch but also from galaxies displaced in redshift.

Adruv is instructed to prove the relic is still functional by commanding it to collect exactly 10 photons that were originally emitted by a Sun-like star at a cosmic redshift $z = 0.2$. The "time machine" claims to be sensitive to every photon arriving in the spectral window between 0.6 and 28.5 μm (from deep orange through mid-infrared), with no losses, and the star can be treated as an undisturbed solar twin that has been transmitting steadily since antiquity.

Unfortunately, no one knows how long Adruv must wait for these 10 photons to be detected. Therefore, estimate numerically the time required for the device to accumulate the 10 photons.

3 Apparently Invisible

The archives of the Interstellar Cartographers Guild whisper of a system long forgotten, nicknamed *Apparently Invisible*. In the black ledger of explorers it is said:

“A dim star of magnitude 7 drifts 385 light-years from Sol. A hidden wanderer circles it, one full revolution every 570 years, always keeping to a perfect ring. The orbit is unbroken, silent, yet the planet’s separation is so slight that even the sharpest eyes of the ancients could not see it.”

Centuries later, a new council stands before the James Webb Space Telescope, which has a mirror of diameter 6.5 m and reaches out to $3.8\text{ }\mu\text{m}$, seeking to resolve the unseen. They argue in endless loops: *Can such a telescope truly split the light of the planet from its star, or will the world remain invisible forever?*

The question is not written clearly in their scrolls. All that remains is this fragment:

“Find the greatest angular parting of the planet from its star, as viewed from Earth, and judge it against the sharpness of Webb’s eye. Decide whether the hidden world may ever be glimpsed.”

Using only the information above, **estimate the maximum angular distance between the planet and its star as it would appear to an observer on Earth. (in arcsecond)**

4 The Martian Encounter

A joint Earth–Mars mission launches a probe designed to slingshot around Mars and head into the outer Solar System. The spacecraft departs Earth on a carefully chosen trajectory: it is accelerated until its heliocentric path becomes a **parabola grazing Earth’s orbit at perihelion**, with the Sun at the focus.

Several months later, the spacecraft crosses the orbit of Mars, just as Mars happens to be near the intersection point. At this instant, imagine you are an observer stationed on Mars, watching the probe approach *before Mars’ gravity significantly alters its path*.

From your Martian vantage point:

- The probe is moving along its parabolic orbit under the Sun’s gravity alone.
- Its heliocentric speed is determined by the orbital mechanics of a parabolic trajectory.

- The trajectory cuts Mars' circular orbit at a nonzero angle, which can be found from the geometry of the orbits.

Determine the probe's **heliocentric approach speed** as it crosses Mars' orbit, and express your answer numerically in kilometres per second (km/s).

5 The Moontide Anomaly

In the year 2457, Earth's oceans are reported to exhibit strange “tidal pulses” whenever the Moon passes overhead. An observer at the equator, call him *Adruv*, notices that these pulses do not follow the classical Newtonian prediction, but instead appear to be shifted in longitude by a strange angle ϕ .

At the same time, the Earth's equatorial radius has increased slightly due to terraforming attempts, now measured as

$$R = 6.46 \times 10^6 \text{ m},$$

while the Earth–Moon distance has shrunk mysteriously to

$$r = 3.51 \times 10^8 \text{ m}.$$

When the Moon is exactly above point B on the equator, *Adruv* is standing at point A , also on the equator but at longitude difference $\phi = 7.5^\circ$ east of B . He reports a strange sideways tug in the water surface beneath him, as if the ocean were trying to “crawl” toward B .

Physicists of that era define this tug as the **horizontal tidal acceleration** of the seawater at A due to the Moon's gravitational field.

Using only the information above (and the constants

$$G = 6.674 \times 10^{-11} \text{ SI}, \quad M_{\text{Moon}} = 7.35 \times 10^{22} \text{ kg}),$$

determine numerically the magnitude of this **horizontal tidal acceleration** (in nm/s²) felt by *Adruv*.

6 Hot Potato

Long ago, deep-space navigators wrote of a swollen blue-white sun they called *The Ember*. Its size was said to be 3.9 times the Sun's radius, its surface burning at 9520 K, and its distance so vast that its parallax seen from Earth was only 0.011″.

Circling *The Ember* is a restless shard of rock, an asteroid that whirls once around the star in exactly one Earth-year. Its skin spins too quickly to ever keep one side warm, and it

throws back 30% of the light that strikes it. For this problem assume the asteroid's radius is

$$r = 1000 \text{ km.}$$

Using only the information above (and standard physical constants), determine:

the asteroid's apparent **bolometric** magnitude as seen from Earth.

(If helpful, you may assume solar values $R_{\odot} = 6.96 \times 10^8 \text{ m}$, $L_{\odot} = 3.828 \times 10^{26} \text{ W}$, $M_{\odot} = 1.9885 \times 10^{30} \text{ kg}$, and $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.)

7 Hide and Seek

The star-navigators of the old colony ships left behind fragments of coded sky-logs. One fragment, recovered from a derelict archive orbiting Jupiter, speaks of a “Hidden Flame” — a beacon star brighter than $+1.5^{\text{m}}$, used to align their engines with the spiral arm.

The record says:

At the 50th northern parallel, the Flame will rise at the exact heartbeat when Antares vanishes beneath the edge of the world, and it will set at the instant Sirius claws its way above the horizon.

The log ends abruptly, but another scrap adds:

“The Flame guides not by Earthly maps, but by its place in the Galaxy itself.”

The Earth is to be considered perfectly spherical, with no air to bend the light.

Question. What is the value of b in the *galactic coordinates* (l, b) of this Hidden Flame? (in degree)

8 One Thousand and One Nights: The Precessing Archive

A battered probe recovered from the Oort Archive carries three fragments of a ruined star chart. Each fragment lists a point on the planet's ecliptic in local equatorial coordinates (angles in degrees):

$$P_1 : (\alpha_1 = 20.4^\circ, \delta_1 = +22.5^\circ), \quad P_2 : (\alpha_2 = 74.7^\circ, \delta_2 = +49.0^\circ), \quad P_3 : (\alpha_3 = 68.1^\circ, \delta_3 = +34.9^\circ).$$

The chart is corrupted: the right ascension zero-point is shifted by an unknown constant yaw of up to $\pm 0.3^\circ$, and each measurement carries random errors of about $\pm 0.2^\circ$. The planet's geometry otherwise mimics Earth's — equator and ecliptic defined in the same way.

Ancient notes add one more curse: the spin axis of this world does not hold still, but precesses, and its obliquity oscillates with amplitude $\Delta\varepsilon = 5.0^\circ$ around the mean obliquity ε that you must first recover from the chart.

Finally, Determine the *Polar-Night Coverage Factor* F_{ever} , the fraction of the planet's surface that will experience polar night at least once during a full precession cycle. Express your final answer as a percentage with two significant figures.

9 Celestial Pause

In a forgotten observatory near the twilight edge of the old world, an astronomer watches a wandering speck of light that seems to hesitate in the heavens. “It stops,” she writes, “as if the sky itself holds its breath.”

The records say this speck — a modest asteroid, endlessly circling in the ecliptic — grows slightly smaller as the days pass after opposition. From that perfect alignment, when Sun, Earth, and the asteroid lie in a straight cosmic line, its apparent disk shrinks by about ten percent before the strange pause occurs — the stationary point, where its drifting path upon the celestial sphere seems to reverse.

No other details are left, save that the observer stood upon the Earth itself, and that the asteroid's path was a flawless circle, lying exactly in the ecliptic plane.

Somewhere in this hesitant motion lies the secret number the astronomer sought: **the radius of the asteroid's orbit. (in AU)**

10 Winter Ensemble

On a frost-bitten night beneath the pale shimmer of Orion, an old observer gazes upon three familiar lights that weave a triangle across the southern sky — Sirius, Procyon, and Betelgeuse. Their places in the heavens are known:

Star	Right Ascension	Declination
Sirius	6 ^h 45 ^m	−16° 45′
Procyon	7 ^h 40 ^m	+05° 10′
Betelgeuse	5 ^h 56 ^m	+07° 24′

As the hours drift toward midnight, the observer wonders: *Is there a place on Earth where these three bright stars might all stand at the same altitude above the horizon at the same instant?*

Therefore, Determine the latitude at which Sirius, Procyon, and Betelgeuse can be at the same altitude at the same time.

Note: Express the latitude numerically in **degrees only**, without using any sign symbols.

11 The Aberrant Sun

In the silent archives of the Deep Sky Survey, astronomers speak of a lone G2V star that seems to waver upon the firmament. Through the seasons it sketches a tiny ellipse, not by its own motion, but by the trembling of Earth's orbit around the Sun. The ancient logbooks note that this aberrational path has an eccentricity of 0.4, and that the star's parallax is merely 1% of the greatest aberrational displacement ever seen.

They whisper that it shines like our own Sun, though far enough that its disk would vanish to almost nothing. Yet one question remains from those faded observations:

From these clues, estimate the *surface brightness* of the stellar disk. Express the result in mag/arcsec².

12 Tides on a Rod

An experimental gradiometer satellite consists of a thin, rigid, massless rod of length $L = 40$ m with small masses $m = 500$ kg at its ends. Its center moves in a circular orbit at an altitude $h = 400$ km above the Earth's surface. In equilibrium, the rod is oriented radially toward the Earth's center.

Assume that the Earth's gravitational field is the same as that of a point mass:

$$\mathbf{g}(\mathbf{r}) = -\frac{GM_{\oplus}}{r^3} \mathbf{r}.$$

Find a small correction to the satellite's circular orbital speed that comes from the fact that

it isn't a point, but has a finite length L :

$$v \approx \sqrt{\frac{GM_{\oplus}}{R_{\oplus} + h}} \left(1 + [\text{some number}] \frac{L^2}{(R_{\oplus} + h)^2} \right).$$

Note: The answer is very small. Write your final result in the form $x \times 10^{-9}$ and report only the value of x .

13 Glimpse of GLIMPSE

Deep within the dusty plane of the Milky Way, an aging infrared telescope peers toward a faint, reddened cluster known as *GLIMPSE-C01*. Astronomers studying the cluster notice that the tip of its red giant branch (TRGB) appears at an apparent magnitude of $K = 8.7^{\text{m}}$. From archival data, they know that for a similar metallicity, $[\text{Fe}/\text{H}] = -1.5$, the absolute magnitude of the TRGB is $M_K = -6.1^{\text{m}}$ Ivanov2005.

The cluster lies at Galactic coordinates $l = 31^\circ$, $b = 0^\circ$, buried behind thick interstellar dust. According to RiekeLebofsky1985, the ratio of total extinctions in the visual and infrared bands is $A_V/A_K = 9$.

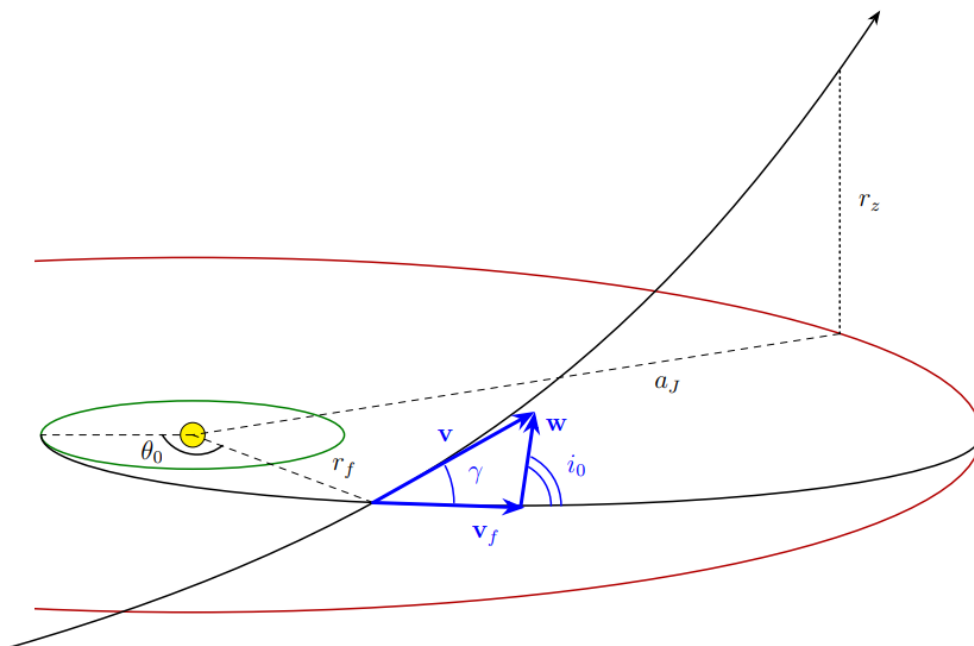
Estimate the total extinction A_V in the V band for GLIMPSE-C01.

14.1 (14) Nothing to Do There

In the year 3421, a lone research vessel called *The Voyager of Silence* drifts through the inner Solar System. Its mission: to awaken the dormant archives buried beneath Jupiter's ancient magnetosphere. Following the commands of its onboard AI, the ship embarks on a Hohmann transfer orbit from Earth's orbit to Jupiter's, gliding in the ecliptic plane under the quiet pull of the Sun. This transfer ellipse just grazes Earth's orbit at perihelion and Jupiter's orbit at aphelion. **(a)** How long would the spacecraft take to travel from Earth to Jupiter along this Hohmann transfer trajectory?

14.2 (15) The Turning of the Flame

Halfway through the long silence, when the craft reaches a true anomaly of $\theta = 90^\circ$ and a heliocentric distance r_f , its ion engines flare briefly — a pulse of plasma as bright as lightning.



The maneuver imparts a sudden velocity change w whose magnitude is $|w| = \frac{1}{2}|v_f|$, where v_f is the spacecraft's instantaneous velocity vector at that point. The thrust is directed at an angle $i_0 = 20^\circ$ out of the ecliptic plane, and its projection onto the ecliptic is aligned with the craft's original velocity.

(b) By what angle γ does the spacecraft's velocity vector turn immediately after the impulse?

14.3 (16) The Orbit of the Shattered Silence

After the burn, the vessel is no longer confined to the ancient plane of the planets. Its trajectory transforms into a new three-dimensional ellipse — a path only the Sun's gravity can define.

(c) Determine the semi-major axis, eccentricity, and inclination of the spacecraft's new orbit after the impulse.

14.4 (17) The Passing Above the Giant

As *The Voyager of Silence* nears Jupiter's orbit, the great planet glows below, unseen, its orbit a thin golden ring in the darkness. The spacecraft passes silently above it, following its new tilted path.

(d) Find the vertical distance r_z by which the spacecraft passes above Jupiter’s orbit.

18.1 (18) Stellar Blends

In the archives of the Galactic Observatory, astronomers recover a faded graph etched on transparent film — the remnant of a long-lost observation. The note beside it reads only: “*Two suns, locked in rhythm, dim and brighten in perfect time.*”

Further examination reveals the data are brightness measurements in the V band of an eclipsing binary star system. The eclipses are central, and both stars are confirmed to be main-sequence objects. One of them, the elder records say, has a spectral type of A0 V.

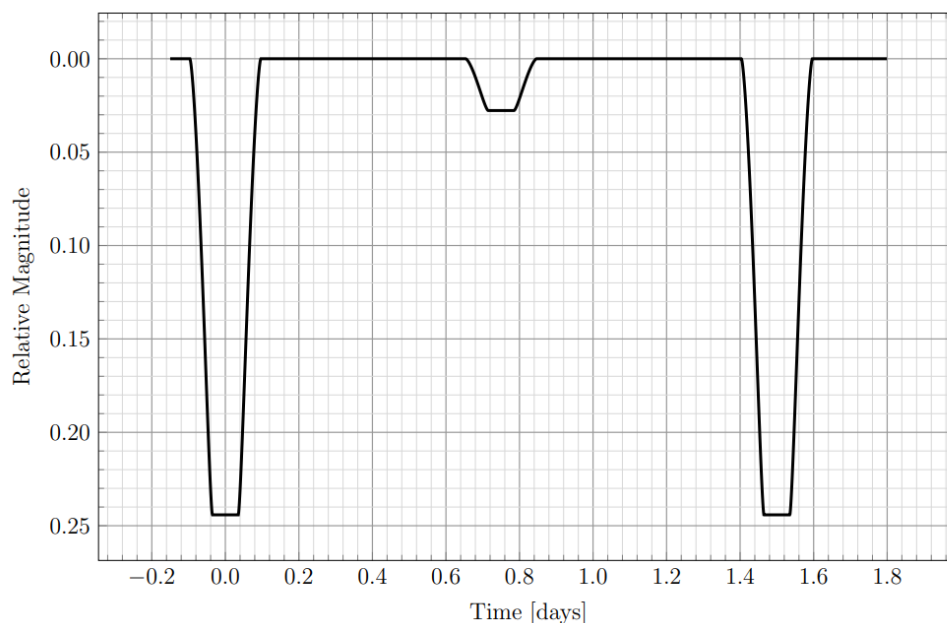


Figure 1: Recovered V -band light curve of the mysterious binary system.

The astronomers debate over the details — what the curve hides in its dips and peaks, and what it whispers about the stars’ hidden geometry.

(a) Determine the ratio of the companion’s radius to the hotter star’s radius,

$$\frac{R_c}{R_h}.$$

18.2 (19) The Distance Between Suns

From the previous question, Estimate the *distance between the two components* of the system, using the orbital properties implied by the observed light curve. Answer in solar radius.

20 Crossroad of Worlds

In the silent archives of an abandoned observatory, two wandering suns — Star A and Star B — are said to drift through the sky, their paths destined to cross more closely with time. Their celestial positions and motions were once inscribed by ancient astronomers, who recorded:

Star	(α, δ)	ϖ (mas)	μ_α (mas/yr)	μ_δ (mas/yr)	V_r (km/s)
A	$(10^{\text{h}}48^{\text{m}}00^{\text{s}}, 0^\circ00)$	80	−160	0	−25
B	$(10^{\text{h}}00^{\text{m}}00^{\text{s}}, 0^\circ00)$	60	+180	0	−20

As the millennia unfold, the stars glide across the firmament, their angular distance slowly shrinking. The astronomer’s riddle asks:

When will the angular distance between Star A and Star B decrease by half, as seen from Earth?

Note. If two possible times are obtained, choose the *larger* one. Express your final answer in **thousand years**.

21 Let the Sky Fall

When the sun rises in the west and sets in the east, when the seas go dry and mountains blow in the wind like leaves... (George R. R. Martin)

In a distant epoch, the Earth’s orbit becomes so distorted that the Sun sometimes rises in the west. At perihelion, the surface grows unbearably hot, yet a remnant atmosphere of molecular nitrogen (N_2) still clings to the planet. Assume the Earth’s mass, radius, and rotation remain unchanged from today.

Estimate the **root-mean-square thermal velocity** v_{rms} of an average air molecule (assume a molar mass of 0.029 kg/mol) when the Sun rises in the west. Consider the spherical albedo of the Earth to be equal to 0.3.

Note. Express your final answer in km s^{-1} .

22 Almost Circumpolar

In a quiet high-latitude observatory, an old astronomer ponders a peculiar question about the dance of the stars. Some of them, he notes, just graze the horizon yet culminate to the

north of the zenith — neither fully circumpolar nor truly setting. He wonders how common such stars might be across the heavens.

Determine what is the maximum possible fraction of such stars and at what latitude this maximum is achieved.

Note: enter four digits (dot considered as a digit, so for example, 2.34 is four digits)

23 Yet Another Pinwheel

From a distant cosmic archive, astronomers recover the spectrum of a faint spiral galaxy — a face-on disc glimmering like a silent pinwheel adrift in the dark. Its structure, they note, follows the familiar exponential law of galactic light:

$$I(R) = I_0 \exp\left(-\frac{R}{h}\right),$$

where R is the distance from the center of the galaxy, h is the radial scale length, and I_0 is the central surface brightness measured in $L_\odot/\text{pc}^2$.

In the recovered spectrum, the center of the H_α line appears at a wavelength of $6670 \text{ \AA}$. The galaxy's apparent magnitude is 14^m , and its scale length is known to be $h = 3 \text{ kpc}$.

Estimate the effective radius R_e of the galaxy. Here, R_e is defined as the radius within which 50% of the total light of the galaxy is emitted. (in Kpc)

24 The Restless Planet

A fast rotating planet of radius R with surface albedo $a = 0.25$ is orbiting a star of luminosity $L = 3.826 \times 10^{26} \text{ W}$. The orbital radius is $D = 1.523 \text{ A.U.}$. It is assumed here that, at equilibrium, all of the energy absorbed by the planet is re-emitted as a blackbody radiation. calculate its surface temperature in kelvins.

25 High Precision Guidance

On a calm morning at the Crimean Astrophysical Observatory, engineers prepare the RT-22 radiotelescope for a delicate test. The dish, a perfect spherical mirror of diameter 22 m and focal length $F = 9.5 \text{ m}$, is to observe the Sun not directly, but in the first sidelobe of its radiation pattern at a frequency of 420 MHz . A sensitive receiver rests at the primary focus of the mirror.

Determine the distance between the **center of the mirror** and the **center of the receiver's shadow** on the dish.

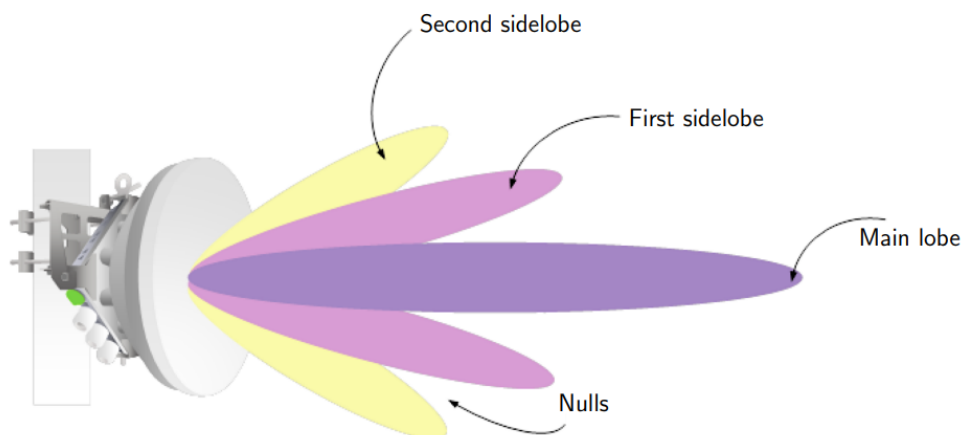


Figure 2: Radiation pattern lobes (beams) of an antenna

Problem 26

The file [Problem 26.csv](#) contains data of two star clusters:

- Plot the HR diagram of both clusters on two different plots (Name the one with more stars cluster X, and the other cluster Y) then on the same plot.
- Which cluster is older? And which one is closer? Why?
- What does the gap between the two cluster main sequence stars represent?
- If the distance to cluster X is 153 ly, what is the distance to cluster Y?

Problem 27

In the field of the globular star cluster **M4**, observed by the *GAIA* space telescope, a number of RR Lyrae variables were found. The equatorial coordinates of the center of M4 are:

$$\alpha = 16^{\text{h}} 23^{\text{m}} 35.22^{\text{s}}, \quad \delta = -26^{\circ} 31' 32.72''.$$

File [Problem 27.csv](#) contains data on the RR Lyrae variables, including:

- the equatorial coordinates (α, δ) of the objects,

- the parallax (in milliarcseconds),
 - the components $\mu_\alpha \cos \delta$ and μ_δ of their proper motion,
 - the period of pulsation P ,
 - the average magnitudes in the G , G_{BP} , and G_{RP} bands,
 - the average radial velocity $\langle V_r \rangle$,
 - the amplitude X_G of pulsation in the G band and its error ΔX_G ,
 - the absorption A_G in the G band.
- (a) Determine which stars do not belong to the cluster.
- (b) Find the period–luminosity relation for RR Lyrae variables of the M4 cluster.
- (c) Estimate the distance to the cluster.
- (d) Estimate the mass of the cluster.

Problem 28

File [Problem 28.csv](#) contains data on some stars of an open cluster associated with a stellar stream:

- the equatorial coordinates (α, δ) of objects,
- the components $(\mu_\alpha \cos \delta, \mu_\delta)$ of their proper motion,
- the radial velocity V_r .

Consider this cluster to consist only of the specified stars.

- (a) Estimate the coordinates of the apex/radiant for the cluster objects.
- (b) Evaluate the parallaxes of individual objects and plot the distribution of parallaxes.
- (c) Determine the statistical parallax of the cluster.

Problem 29

Giza ($29^{\circ}58'N$; $31^{\circ}08'E$) is located on the western bank of the Nile River in Egypt, near Cairo. It is one of the most historically significant archaeological sites in the world and lies close to the Great Pyramid of Khufu.

The file [Problem 29.xlsx](#) shows the lunar eclipses of the 21st century. The table lists the eclipse date, Saros number, γ^1 , the moments of the Moon's disk contacts with the Earth's shadow (see [Problem 29.png](#)), and the time of greatest eclipse.

Point $\Xi(0^{\circ}; 31^{\circ}08'E)$ lies on the equator at the same longitude.

- Find the distance from Giza to the North Pole of the Earth.
- At what solar declination does the polar day begin at Giza? Assume that atmospheric refraction at the horizon is $35'$.
- How many *total* lunar eclipses will occur in the 21st century? *Partial*? *Penumbral*?
- Determine how many *total* lunar eclipses will be visible from both Giza and point Ξ . List the corresponding event numbers.

Count an eclipse as visible if any part of totality occurs above the horizon. Ignore events visible only in partial phases.

Problem 30

The file [Problem 30.csv](#) provides observational data for supernova events. The columns are:

- `sn_name` — supernova designation;
- `redshift` — measured redshift;
- `gal_type` — morphological type of the host galaxy;
- `sn_ra` — right ascension in hours, minutes, and seconds;
- `sn_dec` — declination in degrees, minutes, and seconds.

- Plot the objects on a rectangular sky map with axes (Right Ascension; Declination) in degrees. Are there any regions without objects? What does the largest such region represent?
- For the selected region, plot a histogram of the redshift distribution.

- (c) Keep only objects with redshift $z < 0.1$. For these objects, calculate the pairwise distances l_{ij} for all pairs (i, j) of the remaining objects. The redshift is small — use ordinary Euclidean distance without applying cosmological models.
- (d) Plot a histogram for the distribution $f(l)$ of pairwise distance values.